HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78AA

ACTUARIAL AND FINANCIAL MATHEMATICS A

Semester 1 2017/18

Duration: Two Hours

Total marks: 60

ANSWER ALL QUESTIONS

Electronic calculators approved by the University may be used.

1. (a) A man makes investments annually in advance. The first year's investment is £2 million and the investment increases in each subsequent year by £2 million. Thus the second year's investment is £4 million, the third year's investment is £6 million, and so on for a total of ten years.

Calculate the accumulated value of the investments at the end of the tenth year at a rate of interest of 2% per half-year effective.

[4 marks]

(b) Prove that, if k and n are positive integers, then

$$\ddot{a}_{\overline{n+k}|} = \ddot{a}_{\overline{n}|} + {}_{n|}\ddot{a}_{\overline{k}|}$$

[3 marks]

[Total 7 Marks]

2. An investor purchased a holding of ordinary shares five months before payment of the next dividend is due. Dividends are paid annually and it is expected that the next dividend will be a net amount of 20 pence per share. The investor anticipates that dividends will grow at a constant rate of 6% per annum in perpetuity.

Calculate, in pence, the price per share that the investor should pay to obtain a net return of 8% per annum effective.

[Total 6 Marks]

3. An investor purchases a bond 8 months after its issue. The bond will be redeemed at par ten years after issue and pays coupons of 4% per annum annually in arrears. The investor pays tax at 20% on both income and capital gains.

Calculate the purchase price of the bond per £100 nominal to provide the investor with an after-tax rate of return of 6% per annum effective.

[Total 6 Marks]

4. An investment fund had assets totalling £100 million on 1 January 2015. It received net income of £10 million on 1 January 2016 and £4 million on 1 July 2016. The value of the fund totalled:

- £106 million on 31 December 2015,
- £122 million on 30 June 2016, and
- £135 million on 31 December 2016.

(a) Calculate for the period 1 January 2015 to 31 December 2016, to 5 decimal places:

i. the time-weighted rate of return per annum.

[3 marks]

ii. the linked internal rate of return, using sub-intervals of a calendar year.

[5 marks]

(b) State both in general, and in this particular case, when the linked internal rate of return will be identical to the time-weighted rate of return.

[2 marks]

[Total 10 Marks]

5. A loan of £200 000 was granted on 1 January 2010. The loan is repayable by an annuity payable monthly in arrears for twenty-five years. The amount of the monthly repayment increases by £200 after ten years, and was calculated on the basis of an effective rate of interest of 2% per quarter-year.

(a) Show that the initial amount of monthly repayment is £1 463.64 (minor errors due to rounding will not be penalised).

[7 marks]

(b) Calculate the amount of capital which was repaid on 1 February 2010.

[2 marks]

(c) Calculate the amount of loan which will remain outstanding after the monthly repayment due on 1 January 2017 has been made.

[5 marks]

(d) The borrower requests that after the 1 January 2017 payment has been made, future repayments be a fixed amount payable on each 1 January, the last repayment being on 1 January 2027.

Calculate the amount of the revised annual payment on this basis if the interest rate remains unchanged.

[3 marks]

[Total 17 Marks]

6. (a) An investor is deciding whether to invest in a project. Explain why the discounted payback period is a poorer decision criterion than net present value, on the assumption that the investor is not short of capital.

[2 marks]

An investor is considering two projects, A and B. Project A involves the investment of £1 million at the outset. The only income to be received will be a payment of £3 million after ten years. Project B also involves the investment of £1 million at the outset. Income will be received from this project annually in arrears. In the first year the annual payment will be £0.09 million, in the second year £0.10 million, in the third year £0.11 million, with the annual payment increasing by £0.01 million each year thereafter until the tenth year, after the end of which no further income will be received.

Use a rate of interest of 5% per annum effective for the calculations.

(b) Calculate the net present value of both investment projects.

[7 marks]

(c) Without doing any calculations, state the discounted payback period of Project A.

[1 mark]

(d) Give a reason why the discounted payback period of Project B is before that of Project A (no further calculations are necessary).

[2 marks]

(e) In the light of your answers above, explain which project is the most desirable to an investor with unlimited capital, and why.

[2 marks]

[Total 14 Marks]

END OF PAPER